



Motions & Rates Topic Test 1

Solution

Total Marks: 30

Time: 50 minutes

- 1 The mass of two substances, A and B, are decaying exponentially according to $M_A = 200e^{-0.05t}$ and $M_B = 400 \times 3^{-0.12t}$ where M is the mass in grams and t is the time in minutes.

- (i) Find the time taken for substance A to decrease to half of its original value. Give your answer correct to the nearest minute. 2

As the initial amount of substance A is 200 grams, the time taken to decrease to half its original value is calculated as follows.
Let $M_A = 100$.

$$100 = 200e^{-0.05t} \quad \frac{1}{2} = e^{-0.05t}$$

$$t = \frac{\ln 2}{0.05} = 13.86... \approx 14 \text{ minutes}$$

Therefore, it will decrease to half its original value in 14 minutes.

- (ii) At what instance do both particles decay at the same rate? Give your answer correct to the nearest second. 3

The rate of change of both substances:

$$\frac{dM_A}{dt} = -0.05 \times 200e^{-0.05t} = -10e^{-0.05t}$$

$$\frac{dM_B}{dt} = 400 \times \ln 3 \times -0.12 \times 3^{-0.12t} = -48 \ln 3 \times 3^{-0.12t}$$

Equate the two rates:

$$-10e^{-0.05t} = -48 \ln 3 \times 3^{-0.12t}$$

$$\frac{-10}{-48 \ln 3} = \frac{3^{-0.12t}}{e^{-0.05t}}$$

$$= \frac{e^{(-0.12 \ln 3)t}}{e^{-0.05t}}$$

$$0.1895... = e^{-0.0818t}$$

$$t = \frac{\ln 0.1895...}{-0.0818...} \approx 20 \text{ minutes } 19 \text{ seconds}$$

Therefore, both substances decay at the same rate at 20 minutes and 19 seconds.

- 2 After taking a break at a rest stop, Sergio starts driving down a highway. His car's velocity after t seconds is given by the equation $v(t) = 0.3t^2 \text{ m/s}$.

- (i) Find how long it takes the car's velocity to reach 30 m/s. $0.3t^2 = 30 \quad t^2 = 100 \quad t = 10 \text{ seconds}$ 1
- (ii) Find how far Sergio's car has travelled during this time, in metres. $x(t) = \int_0^{10} (0.3t^2) dt$ 2
- (iii) Find the average speed of the car, in m/s, from when Sergio starts driving down the highway from the rest stop to when it reaches 30 m/s. $\text{average speed} = \frac{100}{10} = 10$ 1

So, the average speed is 10 m/s.

- 3 A particle moves such that its velocity at a given time is given by: $\dot{x} = 8 - 16\sin t$

- (i) What is the initial acceleration of the particle? 1

$$\dot{x} = 8 - 16\sin t \quad \ddot{x} = -16\cos t$$

Initially $t = 0$

$$\ddot{x} = -16 \cos(0) = -16$$

$\therefore$ Initially the acceleration is 16 ms² towards the left.

- (ii) When is the particle first at rest? 1

Particle at rest when velocity is zero ($\dot{x} = 0$)

$$\dot{x} = 8 - 16\sin t = 0 \quad 16\sin t = 8$$

$$\sin t = \frac{1}{2} \quad t = \frac{\pi}{6}, \frac{5\pi}{6}, \dots$$

$\therefore$ First at rest after $\frac{\pi}{6}$ seconds.

- (iii) Given that the particle is initially at the origin, find an equation for the displacement 1

$$\dot{x} = 8 - 16\sin t \quad x = 8t + 16\cos t + C$$

(x), at a time t . When $t = 0$ $x = 0$ (Initially at the origin)

$$0 = 8 \times 0 + 16\cos(0) + C \quad C = -16$$

$$\therefore x = 8t + 16\cos t - 16$$

- (iv) Find the distance travelled between the first two times when the particle is at rest. 1

Distance between $\frac{\pi}{6}$ and $\frac{5\pi}{6}$

$$t = \frac{\pi}{6} \quad x = 8 \times \frac{\pi}{6} + 16\cos\frac{\pi}{6} - 16 = \frac{4\pi}{3} + 8\sqrt{3} - 16$$

$$t = \frac{5\pi}{6} \quad x = 8 \times \frac{5\pi}{6} + 16\cos\frac{5\pi}{6} - 16 = \frac{20\pi}{3} - 8\sqrt{3} - 16$$

$$\text{Distance travelled} = \left(\frac{20\pi}{3} - 8\sqrt{3} - 16\right) - \left(\frac{4\pi}{3} + 8\sqrt{3} - 16\right) = \frac{16\pi}{3} - 16\sqrt{3}$$

- 4 A particle is moving in a straight line, starting from the origin. At time t seconds the particle has a displacement of x metres from the origin and a velocity v m/s. The displacement is given by $x = 2t - 3 \log_e(t + 1)$.

- (i) Find an expression for the velocity v . 2

- (ii) Find the initial velocity. 1

- (iii) Find when the particle comes to rest. 1

At rest when $v = 0 \therefore 2 - \frac{3}{t+1} = 0$

$$2 = \frac{3}{t+1}$$

$$2t + 2 = 3$$

$$\therefore t = \frac{1}{2}$$

At rest when $t = \frac{1}{2}$ second

- (iv) Find the distance travelled by the particle in the first three seconds. 3

$t=0 \quad x=0$, $t=\frac{1}{2} \quad x=1-3\log_e 1.5$

$t=3 \quad x=6-3\log_e 4$

$v=0$ at $t=\frac{1}{2}$ $\leftarrow -1\text{m/s}$

$t=\frac{1}{2}$ $t=0$ $t=3$

$$\text{distance} = 2 \times \left| 1 - 3\log_e 1.5 \right| + 6 - 3\log_e 4$$

$$= 2 \times 0.2162 + 1.8411$$

$$= 2.2739 \dots = 2.27 \text{ m (3. sig. figs)}$$

- 5 The concentration of a pesticide in the soil is modelled by the equation given by: $P = P_0 e^{-kt}$ where P_0 and k are constants, and t is the number of days after the application of the pesticide.

- (i) Once in the soil, the half-life of the pesticide is the time taken for its concentration to be 2

reduced to one half of its initial value. If the half-life of the pesticide is 25 days, find the value of k . Answer correct to 2 significant figures.

$$t = 25 \text{ and } P = \frac{1}{2} P_0$$

$$\frac{1}{2} P_0 = P_0 e^{-k \times 25}$$

$$0.5 = e^{-25k}$$

$$\ln 0.5 = -25k$$

$$k = \ln 0.5 \div -25 = 0.0277 \dots \approx 0.028$$

- (ii) What is the percentage decrease in concentration of the pesticide after 80 days? Answer to the nearest whole number.

$$P = P_0 e^{-0.028 \times 80}$$

$$P = 0.1064 \dots P_0$$

- 6 A particle is moving in a straight line with displacement from the origin in metres and t is time measured in seconds. The particle is initially at rest 4 m to the right of the origin.

- (i) Show that the displacement of the particle is given by $x = 4 \sin\left(2t - \frac{\pi}{2}\right) + 8$.

For the displacement function:

$$x = \int 8 \cos\left(2t - \frac{\pi}{2}\right) dt = 8 \left[\frac{1}{2} \sin\left(2t - \frac{\pi}{2}\right) \right] + C = 4 \sin\left(2t - \frac{\pi}{2}\right) + C$$

When $t = 0$, $x = 4$:

$$4 = 4 \sin\left(-\frac{\pi}{2}\right) + C \quad C = 8 \quad \therefore x = 4 \sin\left(2t - \frac{\pi}{2}\right) + 8$$

- (ii) Show that the particle comes to rest again when it is 12 m to the right of the origin and

$$t = \frac{\pi}{2} \text{ s.} \quad \text{When } t = \frac{\pi}{2}: \quad \frac{dx}{dt} = 8 \cos\left(2 \times \frac{\pi}{2} - \frac{\pi}{2}\right) = 0 \text{ m s}^{-1}$$

$$x = 4 \sin\left(2 \times \frac{\pi}{2} - \frac{\pi}{2}\right) + 8 = 12 \text{ m}$$

Therefore, at $t = \frac{\pi}{2}$, the particle is at rest at 12 m to the right of the origin.

- (iii) By finding a function for acceleration, describe the motion of the particle after $\frac{\pi}{2}$ s and before it next comes to rest.

$$\text{For acceleration: } \frac{d^2x}{dt^2} = -16 \sin\left(2t - \frac{\pi}{2}\right)$$

$$\text{When } t = \frac{\pi}{2}: \quad \frac{d^2x}{dt^2} = -16 \sin\left(2 \times \frac{\pi}{2} - \frac{\pi}{2}\right) = -16 \sin\left(\frac{\pi}{2}\right) = -16$$

Therefore, the particle will move towards the left after being stationary at $t = \frac{\pi}{2}$.



Motions & Rates Topic Test 2

Solution

Total Marks: 30

Time: 50 minutes

- 1 The mass of a whale is modelled by $M = 38 - 38.5(1.2)^{-0.44t}$, where M is the mass of the whale in tonnes and t is its age in years.

(i) What is the mass of the whale when its age is 5 years? $M = 38 - 38.5(1.2)^{-0.44t}$ When $t = 5$
 $M = 12.22$ tonnes (correct to 2 dec. places) 1

(ii) At what rate is the mass M increasing when the whale's age is 5 years? 2

$$\frac{dM}{dt} = -0.44 \times -38.5 \times (1.2)^{-0.44t} \times \ln(1.2)$$

$$\text{When } t = 5, \frac{dM}{dt} = 16.94 \times (1.2)^{-2.2} \times \ln(1.2)$$

$$\frac{dM}{dt} = 2.068 \dots \text{tonnes / year}$$

(iii) At what age will the mass of the whale become 25 tonnes? 3

$$25 = 38 - 38.5 \times (1.2)^{-0.44t}$$

$$0.337 \dots = (1.2)^{-0.44t}$$

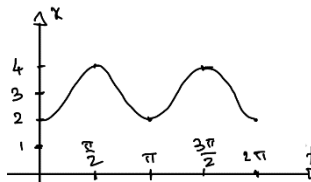
$$\ln(0.337 \dots) = -0.44t \ln(1.2)$$

$$13.53389 \dots = t$$

The mass of the whale will be 25 tonnes after approximately 13.5 years.

- 2 A particle moves in a straight line. At time t seconds its displacement x metres from the origin is given by $x = 3 - \cos 2t$, where $0 \leq t \leq 2\pi$.

(i) Sketch the graph of x as a function of t . 2



(ii) At what times is the particle at rest? 2

Test: $v = 0$
 $\therefore$ stationary point.
 $\therefore t = 0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}, 2\pi$

(iii) Find the time when the particle first reaches its maximum speed after $t = 0$. 2

$$\dot{x} = 2 \sin 2t \quad 2 \sin 2t = 2$$

$$-1 \leq \sin 2t \leq 1 \quad \sin 2t = 1$$

$$-2 \leq 2 \sin 2t \leq 2 \quad 2t = \frac{\pi}{2}$$

$$\therefore -2 \leq \dot{x} \leq 2 \quad \therefore t = \frac{\pi}{4}$$

Max speed = 2.

- 3 A particle is moving in a straight line. At time t seconds it has displacement x metres from a fixed O on the line, velocity v m/s and acceleration $a = 2e^{-t}$ m/s². Initially it is 2 metres to the right of O and is moving towards O with speed 1 m/s.

(i) Show that $v = 1 - 2e^{-t}$ and $x = t + 2e^{-t}$. 2

$$a = 2e^{-t} \quad v = 1 - 2e^{-t}$$

$$v = -2e^{-t} + c \quad x = t + 2e^{-t} + d$$

$$t=0 \left\{ \begin{array}{l} -1 = -2 + c \\ v = -1 \end{array} \right. \Rightarrow \therefore c = 1 \quad t=0 \left\{ \begin{array}{l} 2 = 2 + d \\ x = 2 \end{array} \right. \Rightarrow \therefore d = 0$$

$$\therefore v = 1 - 2e^{-t} \quad \therefore x = t + 2e^{-t}$$

(ii) Find in simplest exact form when and where the particle is at rest. 2

$$v = 0 \quad x = t + 2e^{-t}$$

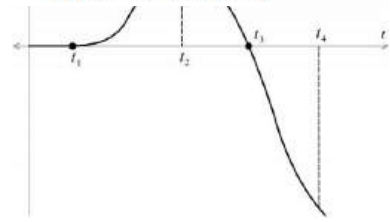
$$e^{-t} = \frac{1}{2} \quad x = \ln 2 + 2 \times \left(\frac{1}{2}\right)$$

$$t = \ln 2 \quad = 1 + \ln 2$$

At rest $(1 + \ln 2)$ m to right of O at time $\ln 2$ seconds.

(iii) Find the distance travelled by the particle in the first 3 seconds of its motion. 2

- 4 A particle moves $t = 3 \rightarrow x = 3 + 2e^{-3}$. Hence in the first seconds
 $(3 + 2e^{-3}) - (1 + \ln 2) = 2 + 2e^{-3} - \ln 2 \approx 1.4 \text{ m}$
 of the origin. The velocity time graph shown below describes this motion.



(i) What is the displacement of the particle at t_1 seconds? 1

10m to the right of origin.

(ii) At what time(s) is the particle at rest? $0 < t < t_1$ and t_3 2

(iii) At what time is the particle farthest to the right of the origin? t_3 1

- 5 The acceleration of a particle, P , in m/s^2 is $\frac{d^2x}{dt^2} = e^{-t} + e^{-2t}$ where t is measured in seconds.

Initially, the displacement of the particle is $x = \frac{3}{4} \text{ m}$, travelling at a velocity $\frac{dx}{dt} = -\frac{3}{2} \text{ m/s}$.

(i) Show that the displacement of the particle is given by: $x = e^{-t} + \frac{1}{4}e^{-2t} - \frac{1}{2}$. 2

$$\begin{aligned} \frac{dx}{dt} &= -e^{-t} - \frac{1}{2}e^{-2t} + c_1 & \text{Initial conditions:} \\ -\frac{3}{2} &= -e^0 - \frac{1}{2}e^0 + c_1 & \rightarrow c_1 = 0 \\ \therefore \frac{dx}{dt} &= -e^{-t} - \frac{1}{2}e^{-2t} & x = e^{-t} + \frac{1}{4}e^{-2t} + c_2 \\ \text{Initial conditions:} \\ \frac{3}{4} &= e^0 + \frac{1}{4}e^0 + c_2 & \rightarrow c_2 = -\frac{1}{2} \\ \therefore x &= e^{-t} + \frac{1}{4}e^{-2t} - \frac{1}{2} & \text{as required.} \end{aligned}$$

(ii) Find the limit of the displacement of P , and hence the limit of the distance that P travels 1

after $t = 0$.

$$\lim_{t \rightarrow \infty} x = 0 + \frac{1}{4}(0) - \frac{1}{2} = -\frac{1}{2}$$

So, the limiting change in displacement
from the starting point is 1.25m.

- 6 The length of daylight, $L(t)$, is defined as the number of hours from sunrise to sunset, and can

be modelled by the equation $L(t) = 12 + \cos\left(\frac{2\pi t}{366}\right)$ where t is the number of days after 21

December 2015, for $0 \leq t \leq 366$.

(i) Find the length of daylight on 21 December 2015. 1

$$L(t) = 12 + \cos(0) = 13 \text{ hours}$$

(ii) What is the shortest length of daylight? 2

$$\begin{aligned} \text{amp} &= 1 & \text{Centre} &= 12 \\ \text{shortest day length} &= 12 - 1 \\ &= 11 \text{ hours} \end{aligned}$$

(iii) What are the two values of t for which the length of daylight is 12? 2

$$\begin{aligned} 12 &= 12 + \cos\left(\frac{2\pi t}{366}\right) \\ 0 &= \cos\left(\frac{2\pi t}{366}\right) \\ \frac{2\pi t}{366} &= \frac{\pi}{2}, \frac{3\pi}{2} \\ t &= \left(\frac{\pi}{2}\right) \div \frac{2\pi}{366}, \quad \frac{3\pi}{2} \div \frac{2\pi}{366} \\ &= 91.5, 274.5 \end{aligned}$$

